

PART II



Sport-Specific Application of High-Intensity Interval Training

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Combat Sports

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Performance Demands of Combat Sports

Combat sports encompass a wide variety of full-contact martial arts and fighting disciplines, many of which have indigenous regional and national forms. Depending on each discipline and the discrete rules of competition, competitors employ strikes with their fists, elbows, knees, shins, or feet (e.g., boxing, karate, taekwondo, and Muay Thai), engage in wrestling and grappling techniques (e.g., judo, sambo, freestyle or Greco-Roman wrestling, and Brazilian jiu-jitsu), or utilize a mixture of fighting styles in a hybrid fashion (e.g., mixed martial arts (MMA), wushu, and hapkido).

Sport Description and Factors of Winning

Combat sports have grown in popularity over the past decade, with MMA now recognized as the fastest growing sport in the world. The global popularity of combat sports is reflected by the Ultimate Fighting Championship (UFC) holding more than 400 events in 133 cities within 22 countries around the world. MMA now exceeds all other combat disciplines in popularity, consistently achieving more pay-per-view viewers than boxing. In 2016, UFC 202 exceeded previous records by attracting 1.65 million pay-per-view buys.

Combat sports have amateur and professional tiers of competition in which men and women compete. The majority involve one-on-one full-contact fighting, however, some combat sports such as Karate and Taekwondo have team competitions that can also be contested. Owing to the different technical, physical, and physiological characteristics found between combat disciplines, competition structures vary greatly. Bouts can comprise anywhere from 3 to 12 rounds, with individual rounds lasting between 2 to 5 min, or be contested during continuous matches ranging from 5 to 30 min. Winning is achieved by a variety of methods, including disabling an opponent at any point during the fight via knockout or submission or by rendering the opponent unable to continue; by an official's intervention to stop the fight; or by outscoring an opponent based on electronic scoring or a judge's decisions (11). The complexities of combat sports mean a fight can be over in an instant or see competitors completing the required number of rounds in their entirety.

Relative Contribution of Physical Performance

The physical demands of combat sports make them among the most complex of all sports, with athletes requiring high levels of metabolic conditioning

concurrently with the ability to generate explosive knockout strength and power. While the characteristics of individual fighting disciplines vary, combat sports are collectively defined as high-intensity intermittent exercise in which force must be repeatedly exerted against an external resistance in the form of an opponent. High-force-generating qualities are required to manipulate the mass of an opponent, withstand collisions, or underpin high-velocity techniques such as strikes, throws, and takedowns (30). These actions must be expressed concurrently with levels of metabolic conditioning that meet the energetic requirements of each particular discipline. Attempts have been made to quantify workload in combat sports, but it remains difficult to clearly define the physiological requirements of competition. Instead, effort:pause ratios during each round of a fight have been used as a proxy of physiological load, with judo, wrestling, and karate reported to have typical effort:pause ratios of 2:1 (5, 43), boxing is approximately 3:1 (34), taekwondo ranges from 1:3 to 1:6 (25, 40), Brazilian jiu-jitsu matches are near to 10:1 (51), and MMA ranges from 1:2 in regional events, and up to 1:4 in the UFC (18, 41) (see figure 11.1).

Targets of Physical Performance in Combat Sports

Effort:pause characteristics can be used to define workload distribution, and from these insights appropriate training strategies can be developed. This can be particularly challenging for hybrid combat sports, however, as the large combination of techniques necessary to succeed places great demand on divergent physiology (e.g., explosive striking versus sustained grappling). Figure 11.2 shows the relative importance of the three main physical qualities of speed, strength, and endurance for elite participation in combat sports and the general relative importance of skills, tactical awareness, and physical capacities.

Combat athletes must also consider the importance of body composition and weight management, as most competitions have weight classifications in which athletes are matched by body weight. The majority of combat athletes therefore attempt to maximize lean tissue and minimize body fat, all while minimizing total body weight and maintaining optimal performance standards (53). When pooling all weight classes together, unpublished data in professional UFC

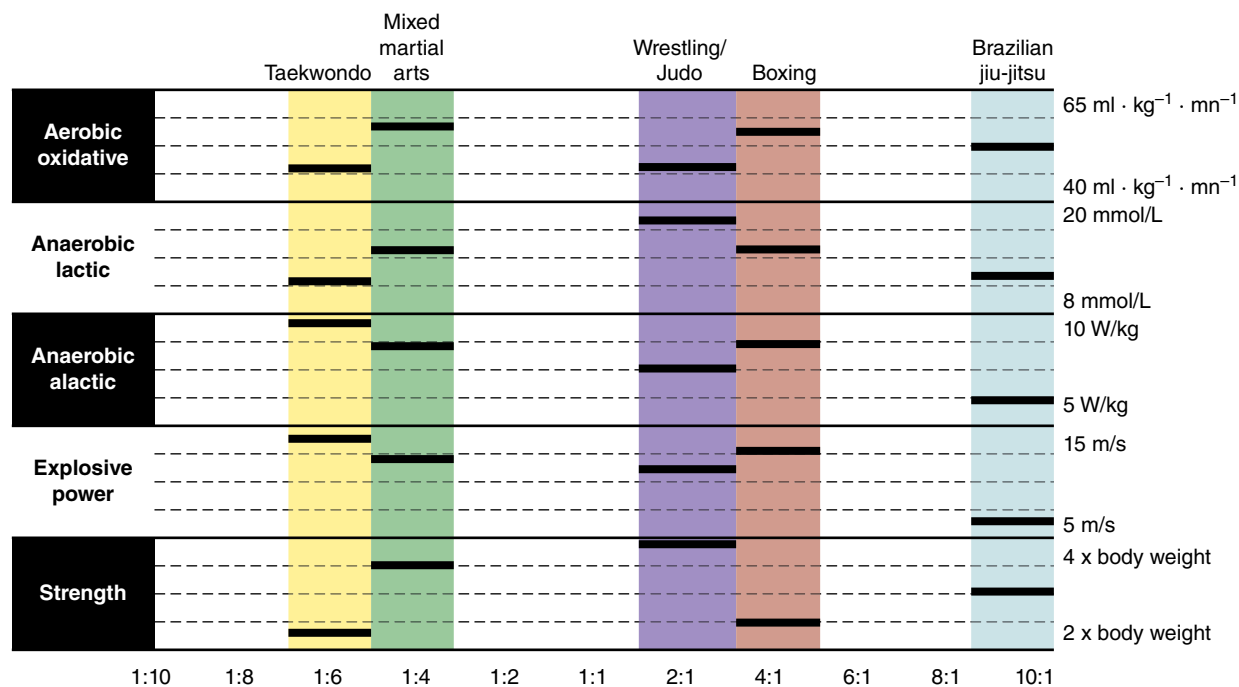


Figure 11.1 Schematic representation of the physiological attributes associated with combat sport disciplines, N.B. data not to scale.

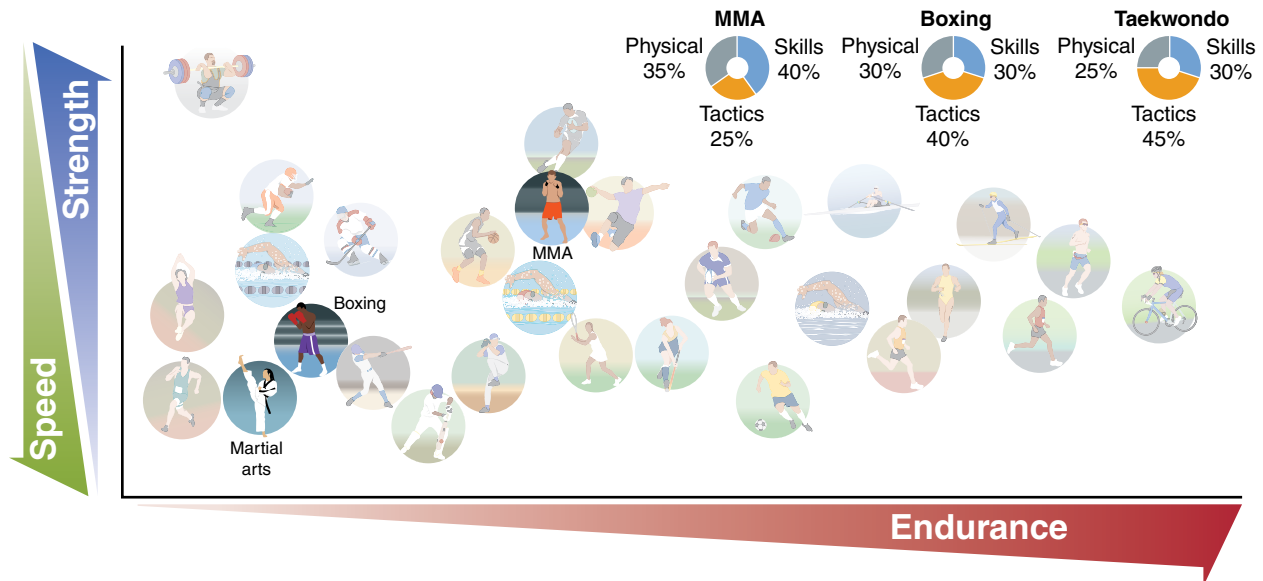


Figure 11.2 The position of the combat sport athlete on the three axes illustrates the relative importance of the three main physical qualities of speed, strength, and endurance for elite participation, acknowledging this varies between categories. Outside of critical psychological components, which are difficult to quantify, the pie charts show the general relative importance of skills, tactical awareness, and physical capacities for combat sport athlete success.

Adapted from G.A. Nader, "Concurrent Strength and Endurance Training: From Molecules to Man," *Medicine & Science in Sports & Exercise* 38, no. 11 (2006): 1965-1970.

fighters indicates athletes vary between an average of 19.1% body fat when off camp, to 17.3% early in camp, 14.0% midway through a fight camp, and are approximately 13.3% at the latter stages of a fight camp when measured by dual-energy X-ray absorptiometry. The following section describes the physiological demands of each combat sport category in greater detail.

Striking

Striking sports, such as boxing, karate, taekwondo, and Muay Thai, involve highly complex motions of the arms or legs to inflict physical damage on an opponent. Striking techniques including punches, elbows, knees, and kicks are extremely dynamic in nature and occur in a fraction of a second. Primarily a function of the high-energy phosphate system, mean impact velocities range from 8.9 to 9.14 m/s for a boxing punch (4, 39, 52) to 7.3 to 14.4 m/s for karate, taekwondo, and kung-fu kicks (19, 45). Anaerobic energy production is critical for striking disciplines, with the high-energy phosphate (i.e., ATP-PCr) and glycolytic systems providing the primary sources of energy for high-impulse movements. Peak (~9.3 W/kg) and mean (~6.7 W/kg) anaerobic power is similar among the striking sports

of boxing (14), taekwondo (7), and karate (13), but somewhat lower than standards found in wrestlers (53). Oxidative energy production is also an important consideration in striking sports (17, 48), as aerobic metabolism supports the athlete's ability to maintain repetitive high-intensity actions during a bout, accelerate the recovery processes between discrete high-intensity efforts, and fulfill chronic work rates throughout the duration of a fight. Interestingly, mean aerobic power is comparable between all striking disciplines, with the value of 44 to 65 ml·kg⁻¹·min⁻¹ reported for boxers similar to that found in karate (47 to 61 ml·kg⁻¹·min⁻¹) (13) and taekwondo (44 to 63 ml·kg⁻¹·min⁻¹) fighters (7).

Wrestling

Wrestling techniques are used in judo, sambo, free-style or Greco-Roman wrestling, wushu, and MMA. To win in wrestling you must attempt to pin your opponent by forcing both shoulder blades to touch the mat using maneuvers such as takedowns, reversals, and escapes. Like all combat sports, the physiological requirements of wrestling are challenging, as matches can last a matter of seconds or as long as 9 min, possibly stressing explosiveness of the lower

body or muscular endurance of the upper body. Having greater absolute strength than an opponent, or by achieving greater relative strength based on weight class, can give wrestlers a competitive advantage in a sport in which manipulating the body weight and resistance of an opponent is critical. In comparisons of successful versus less successful wrestlers, absolute strength levels are greater in more successful athletes, particularly in the upper body (49).

Energy expenditure during a wrestling match has been estimated to be 13 to 14 kcal/min, giving wrestling one of the highest rates of expenditure among all sports (26). Anaerobic peak power and mean power (i.e., anaerobic capacity) are critical physical attributes associated with the explosive high-impulse maneuvers that lead to control of an opponent. The maintenance of anaerobic power for the duration of a bout is essential in wrestling and is likely due to an athlete's ability to utilize and replenish anaerobic glycolytic capacity and buffer metabolic acids. In junior wrestlers, anaerobic power and capacity are as much as 13% higher in elite performers compared to recreational age- and bodyweight-matched opponents (27). Elite Korean wrestlers have been shown to have venous lactate concentrations of 10 to 13 mmol/L following 5 min of wrestling competition (54), while Kraemer et al. (35) reported concentrations as high as 20 mmol/L in NCAA collegiate wrestlers. When comparing aerobic power (i.e., $\dot{V}O_{2\text{peak}}$) in wrestlers, both successful and less successful wrestlers from Olympic, collegiate, and scholastic populations have similar values of 53 to 56 ml·kg⁻¹·min⁻¹ (27, 53). Owing to the complex energetic demands of wrestling, effective high-intensity interval training (HIIT) modalities are an essential way to augment both the aerobic and anaerobic systems that regulate a wrestler's ability to sustain high-intensity workloads.

Grappling

Grappling techniques can be used in judo, sambo, freestyle or Greco-Roman wrestling, Brazilian jiu-jitsu, and MMA. Instead of trying to pin an opponent, grappling uses submission holds (e.g., chokes, arm bars, and leg locks) to place opponents in positions that cause extreme pain, fear of injury, or loss of consciousness, thereby forcing them to submit out of a match. In comparison to wrestling, in some forms of grappling, lying on your back underneath your opponent can actually be advantageous, as it

gives fighters the opportunity to hold their opponent with their legs and perform submission holds. A typical jiu-jitsu grappling match comprises intervals of ~170 s of activity, of which 25 ± 7 s is standing combat and 146 ± 119 s is groundwork, interspersed with 13 ± 6 s intervals of active rest (51).

Elite-level jiu-jitsu fighters have excellent levels of abdominal and upper-body strength endurance (3). These same athletes have moderate to low levels of maximal leg strength, perhaps indicative that grappling relies more on ground-based chokes and submission holds rather than the explosive throwing movements normally seen in judo and wrestling. Brazilian jiu-jitsu fighters have limited maximal isometric strength levels in comparison to judo and wrestling athletes, but they are able to maintain sustained levels of strength throughout a fight, with upper-body strength endurance a critical attribute for grappling disciplines (44). Metabolically, grappling elevates several biochemical markers relating to the high physiological stress induced by intense exercise, as well as severe microtrauma to muscle tissue (6). Aerobic power (i.e., $\dot{V}O_{2\text{max}}$) in elite Brazilian jiu-jitsu fighters is approximately 50 ml·kg⁻¹·min⁻¹, somewhat lower than the values seen in both judo athletes (12, 20) and wrestlers (26, 53). These findings most likely reflect that the metabolic cost is diminished somewhat by athletes lying semipassively on their backs for sustained periods of activity. The specific characteristics of grappling influence the need to select appropriate training modalities, with HIIT focusing less on explosive force expression and more on the need for sustained work output for longer durations.

Hybrid

Combining all combat modalities, hybrid combat sports such as MMA, wushu, and hapkido utilize a mixture of striking, wrestling, and grappling techniques. Hybrid disciplines distinguish themselves by the highly complex skillsets athletes must possess. Victory is not only influenced by a fighter's ability to land strikes (e.g., Muay Thai) but also by his proficiency in defending against or successfully achieving takedowns (e.g., wrestling) and then executing ground fighting techniques (e.g., grappling). These complex requirements place significant demand on divergent physiology, making training for hybrid sports very complex and multifactorial (figure 11.1). In professional MMA, high-intensity epochs of activity occur for approximately 8 to 10 s, interspersed

with periods of lower-intensity activity (e.g., clinch work, grappling) lasting 2 to 3 times as long (41). Notably, 77% of all MMA contests are ended during high-intensity phases of activity (18). The challenging performance dynamics of hybrid combat sports lend themselves to intermittent HIIT strategies, as the need to develop the physiological attributes that support both explosive force production (i.e., striking) and prolonged work output (i.e., wrestling/grappling) can be effectively distributed throughout a training phase using this approach.

High-level MMA fighters possess greater upper- and lower-body strength levels than lower-level fighters (31). When comparing hybrid athletes from MMA with other combat disciplines, MMA fighters are most physiologically similar to judo players and least like kung fu athletes (47). Del Vecchio et al. (18) report that approximately twice as many MMA fights end during high-intensity groundwork sequences (50%) than intense striking exchanges (26.9%), further reinforcing the need for high levels of muscular strength to support grappling techniques. However, striking is an essential part of MMA, particularly when an athlete has achieved an advantageous top position (e.g., full mount). Indeed, 80.4% of fighters who record four or more significant strikes per bout while in a dominant position on the ground ultimately go on to win the fight (32). Significant ground strikes landed per minute, total strikes landed per minute, and striking accuracy all contribute to success in MMA (32).

Hybrid combat sports place demand on both anaerobic and aerobic energy pathways (figure 11.1). With greater than three-quarters (i.e., 77%) of MMA fights ending during high-intensity episodes lasting only a few seconds (18), there is a need for athletes to have a well-developed neuromuscular system that support strikes, throws, and takedowns. Anaerobic supply of ATP accounts for 90% of energy contribution for maximal efforts lasting ~10 to 15 s (21). However, repeatedly executing high-intensity episodes also challenges glycolytic capacity and the ability to tolerate extremely high metabolic acidosis. Lactate levels up to 20 mmol/L are reported following both MMA training and competition (2). Because of repeated high-intensity effort, it is likely that even greater lengths of time are spent operating near maximal aerobic power (i.e., >90% $\dot{V}O_{2peak}$). Hybrid combat sports recruit type II muscle fibers in combination with high levels of cardiac output, which is ultimately driven by increased stroke volume and associated myocardium enlarge-

ment (36). Superior aerobic power is therefore a critical physiological parameter, with MMA fighters at a regional level reported to have $\dot{V}O_{2peak}$ values of $55.5 \pm 7.3 \text{ ml}\cdot\text{kg}^{-1}\cdot\text{min}^{-1}$ (47) and professional fighters in the UFC as high as $62.75 \pm 4.8 \text{ ml}\cdot\text{kg}^{-1}\cdot\text{min}^{-1}$ (1). The variety of ways in which HIIT can be manipulated to target specific physiological (i.e., high-energy phosphates, anaerobic glycolysis, or oxidative phosphorylation) and neuromuscular systems means it represents a valuable approach to the physical preparation of hybrid fighters.

Key Weapons, Manipulations, and Surveillance Tools

The complexity of combat sport physiology means a variety of HIIT methods are required to adequately target both the metabolic and neuromuscular characteristics that determine success. This section discusses some of the HIIT approaches used in combat sports, the distribution of which is outlined in figure 11.3.

HIIT Weapons

Aerobic, anaerobic lactic, and anaerobic alactic contributions depend on the changing intensities and duration of each combat discipline. To develop the physiological attributes required for performance in combat sport, metabolic conditioning must address

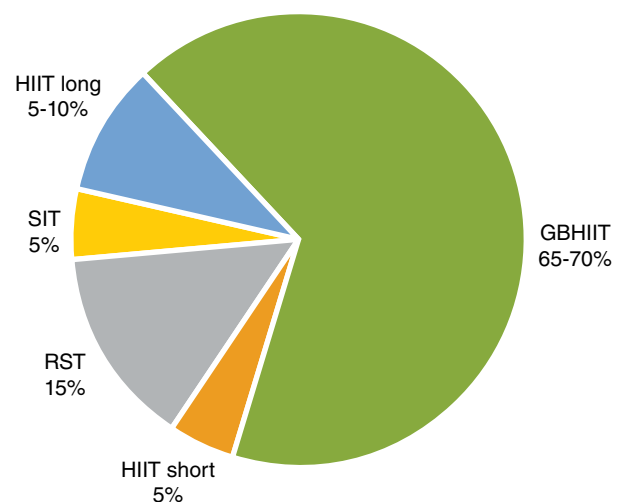


Figure 11.3 Percentage of the different HIIT weapons (formats) used throughout the annual season in combat sports. HIIT: high-intensity interval training; RST: repeated-sprint training; SIT: sprint interval training; GBHIIT: game-based HIIT in the form of fight-based HIIT (FBHIIT).

the five key bioenergetic and neuromuscular load targets (figure 1.5):

1. Aerobic oxidative (HIIT type 1)
2. Aerobic oxidative + neuromuscular (HIIT type 2)
3. Aerobic oxidative + anaerobic glycolytic (HIIT type 3)
4. Aerobic oxidative + anaerobic glycolytic + neuromuscular (HIIT type 4)
5. Anaerobic glycolytic + neuromuscular (HIIT type 5)

The most effective training programs coordinate all aspects of fight preparation and skill training with supplementary strength and conditioning activities (type 6 response; chapter 6). The distribution of each load target (i.e., HIIT types 1 to 5) within a training plan is critical to achieving the desired adaptations, with combat sports placing great demand on all respective metabolic and neuromuscular systems. It is important to realize that HIIT induces distinct effects on all systems of the body, and therefore the way HIIT is distributed between training phases is fundamental in determining the adaptations that can be achieved.

Manipulations of Interval Training Variables

Manipulating HIIT variables must first be observed at the most fundamental level of training variation (i.e., periodization). Based on the specific combat discipline and the level of participants, athletes will either have a defined competition structure (e.g., scheduled open tournaments, national or world championships, Olympics; see figure 11.4) or be randomly assigned to fight bouts (e.g., contractually signed to professional fight cards) throughout the year. Generally, fighters can be considered either in out-of-competition or off-camp (≥ 8 to 12 wk from their next bout) phase or in a fight preparation phase or fight camp (normally ≤ 8 to 6 wk prior to the next fight or tournament). In simple terms, the primary goal of out-of-competition conditioning is to augment *maximal energy production* capabilities (i.e., metabolic power), while fight camps should focus on developing the *efficiency to utilize energy* (i.e., metabolic efficiency) (33).

HIIT With Long Intervals

Organized into multiple rounds with intermittent rest periods, many combat sports are ideally suited

to long training intervals. Long-interval HIIT develops both aerobic and anaerobic energy pathways (9, 10) and impacts bioenergetic target types 3 and 4. In combat sports, long intervals are typically performed between 90 s to 5 min at 80 to 95% HR_{peak} , and are often used to replicate specific competition structures (e.g., 2 min rounds for taekwondo, 3 min rounds for boxing, 5 min rounds in MMA). Heart rate zones used in conjunction with ratings of perceived exertion (RPE) are a valuable tool to gain insight to training intensity for combat sports, as reference speeds like maximal aerobic speed (MAS) for running is rarely used. As most combat training strategies do not involve running, they therefore must be interpreted with caution, as certain modalities may challenge physiological responses in different ways (e.g., the effort required to achieve 80% to 95% HR_{peak} via grappling versus striking bags). Some combat categories (e.g., wrestling, grappling) do, however, adopt extremely long work intervals of up to 15 min continuous activity during out-of-competition training phases to overload both muscular strength and endurance. Long-interval strategies are easily applied to a variety of sport-specific drills, as well as more generic strength and conditioning activities such as metabolic circuits and machine-based intervals (see table 11.1).

The crossover between anaerobic and oxidative pathways occurs at ~2 min of the first interval, meaning that during 2 min intervals approximately 50% of the total ATP during the first interval may be derived from anaerobic pathways and 50% from aerobic pathways (figure 11.5). Over a 4 min effort, approximately 60% would be derived aerobically and 40% anaerobically; in 6 mins, 75% would be aerobic and 25% anaerobic, and so on (38). Metabolic limitations to long-interval HIIT are therefore mechanisms that limit anaerobic power (e.g., glycolytic enzyme kinetics) and capacity (e.g., H^+ tolerance and buffering capacity), as well as those that limit maximal aerobic power (e.g., oxidative enzyme kinetics, stroke volume, myocardial enlargement; chapter 3). Using long-interval HIIT to develop anaerobic capacity as well as the oxidative system not only improves energy supply but also aids in processing the removal of metabolic by-products and the replenishment of muscle glycolytic capacity. Without a well-developed aerobic system, a fighter's anaerobic capabilities are essentially limited, as it will ultimately take longer before he or she is able to produce alactic and lactic energy again to enable the critical combat

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Figure 11.4 Sample combat sport (taekwondo) macrocycle for Olympic Games preparation. The distribution of phases (i.e., out of competition and fight camp) throughout the year is based on a set competition schedule. Phases are further divided into general, specific, and competition periods to further define the focus of preparatory activities throughout the year and peak performance standards for competition. Note, competitions are identified as primary (black) or secondary (white) based on the amount of ranking points available, with the major focus of the year (i.e., Olympic Games) shown in gold.

Table 11.1 Combat Sport-Specific High-Intensity Interval Training (HIIT) Workouts With Long Intervals

These long-interval (approximately 90 s to 5 min at 80% to 100% HR_{peak}) HIIT workouts focus on the development of physiological attributes associated with bioenergetic target types 3 and 4.

	Striking (boxing)	Wrestling (freestyle)	Grappling (BJJ)	Hybrid (MMA)
Target	↑ oxidative capabilities of ST and FT fibers	↑ cardiac output, ↑ mitochondrial proliferation, and ↑ oxygen utilization	↑ maximum rate of oxygen delivery and the ability of FT fibers to utilize oxidative metabolism	↑ maximum rate of oxygen delivery and oxidative capabilities of ST and FT fibers
HIIT type	4	3	3	4
Modality	Boxing the heavy bag	Shadow wrestling	Technical drilling	Metabolic circuit (auxiliary strength and conditioning)
Format	Simulated boxing rounds using various punching techniques to hit the heavy bag	Shadow wrestle by moving in all directions, intermittently simulating double-leg or high crotch takedowns or sprawl defense. Focus on stance and movement.	Semicompetitive drilling with sustained defensive pressure from partner.	Circuit of five auxiliary exercises that challenge force expression and max work rates (e.g., VersaClimber, battle ropes, Airdyne, strong-man variants, throws)
Intensity	Moderately high (80% to 90% HR_{max})	Moderate (70% to 85% HR_{max})	Moderately high (80% to 95% HR_{max})	High (85% to 100% HR_{max})
Volume	12 × 3 min	10 × 2 min	3 × 8 min	2 × (3 × 5 min), 3 min between sets
Work:Rest	3:1	3:1	2:1	5:1

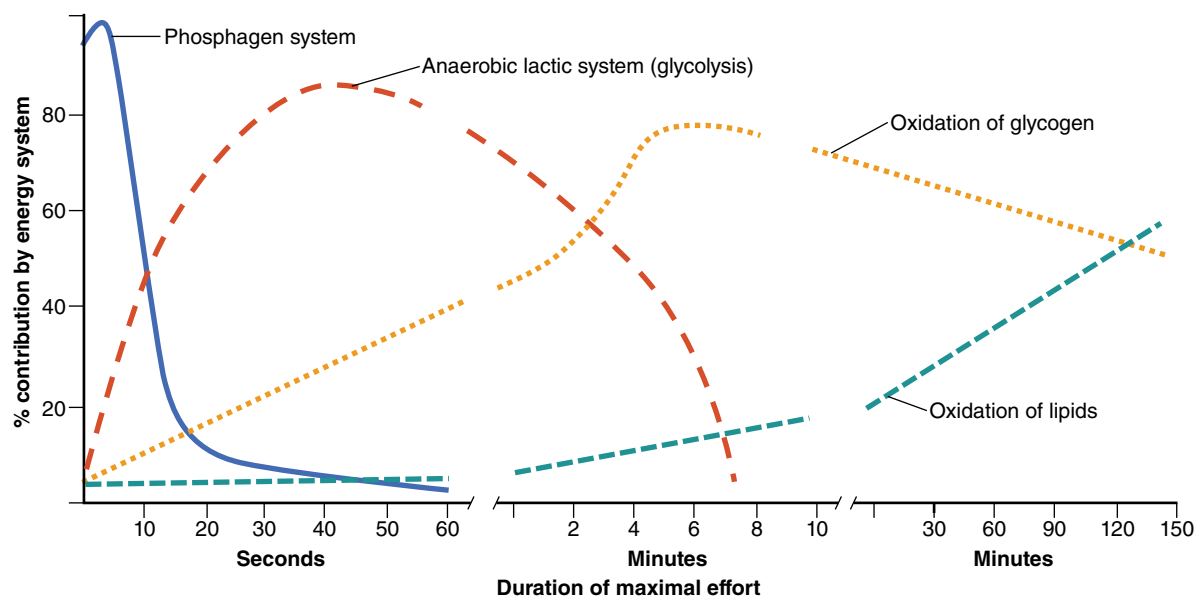


Figure 11.5 Relative contribution of the four energy-delivery systems according to the duration of the event or working interval (time of maximal effort).

maneuvers (33). Long-interval HIIT is primarily used to develop oxidative power during out-of-competition training and during the early stages of a fight camp where *metabolic efficiency* is the priority.

HIIT With Short Intervals

Short intervals in combat sport training are primarily used to develop the high-energy phosphate (i.e., alactic) and lactic systems, which impact bioenergetic target types 2, 3, and 4. In combat sports, short-interval HIIT ranges from repeated single near-maximal-intensity efforts (i.e., type 2) that stress alactic neuromuscular power (e.g., kickboxing strikes or medicine ball throws) to intervals of 10 sec-on/10 sec-off up to 40 sec-on/180 sec-off (i.e., types 3 and 4), which develop lactic power (e.g., takedown drills or clinch defense). High-energy phosphates are critical to the high-impulse movements required in combat sport. Short intervals increase enzymatic activities that facilitate alactic energy production as well as increase the stores of readily available phosphocreatine within the working muscles (38). Primarily used during fight camps, short intervals

develop metabolic efficiency to support the high-velocity movements essential to striking and take-down maneuvers.

The maximal rate of lactic ATP production can also be increased by short-interval HIIT. Lactic power intervals and explosive repeats can be performed with fight-specific drills (e.g., takedowns, throws, striking, and passing) or generic cardio exercises (33), and are often performed out of competition to augment maximal anaerobic lactic power. As with any training method focusing on increasing power, short-interval HIIT should be performed at $\geq 100\%$ intensity with very high work rates. The majority of combat sports adopt 10 to 40 s bursts of maximal intensity effort, including wrestling takedowns, combination punches, or repeatedly kicking bags (see table 11.2). These drills place great demand on fast-twitch power output during sustained high-velocity movements, and therefore promote energy production via alactic and lactate metabolism (i.e., lactate power). They do, however, induce high levels of metabolic fatigue, and for this reason should be prioritized away from competition.

Table 11.2 Combat Sport-Specific High-Intensity Interval Training (HIIT) Workouts With Short Intervals

These short-interval (approximately 10 to 60 s at 90 to 100% HR_{peak}) HIIT workouts focus on the development of physiological attributes associated with bioenergetic target types 1, 2, 3, and 4.

	Striking (taekwondo)	Wrestling (judo)	Grappling (BJJ)	Hybrid (MMA)
Target	↑ alactic energy production and power output of FT fibers ↑ enzymatic activity and stores of readily available PCr	↑ alactic power ↑ neuromuscular strength and power characteristics	↑ alactic power and power output of FT and ST fibers ↑ oxidative process of recovery	↑ alactic power output of FT fibers ↑ enzymatic activity and stores of readily available PCr
HIIT type	3	2	1	4
Modality	Leg kick combinations against partner's body armor, heavy bag, or double kick pads	Repeat hip throws (O-Goshi technique)	Technical drilling of transitions	Striking to takedown transition drills
Format	Fighters throw high velocity leg kick combinations for a set high-intensity interval, focusing on executing the maximal number of repetitions in the set time.	Athletes complete the maximal amount of hip throws against partners offering passive resistance in the set time	Simulated technique transitions from top position to submission hold (full speed) with semipassive resistance	Work with partner striking Muay-Thai kick/knee combinations then immediately transitioning into takedowns to a dominant top position
Intensity	Near maximal (85% to 100% HR_{max})	Near maximal (85% to 100% HR_{max})	Near maximal (85% to 100% HR_{max})	Near maximal (85% to 100% HR_{max})
Volume	4 × (4 × 15 s), 45 s rest between sets	8 × 30 s	2 × (8 × 20 s), 30 s rest between sets	3 × (30 s striking + ≤ 30 s to achieve takedown × 5), 90 s rest between sets
Work:Rest	1:3	1:4	1:3	1:3

Repeated-Sprint Training

Maximal effort repeat-sprint intervals provide the most effective stimulus to upregulate glycolytic enzymes and increase anaerobic lactic capacity (38). Repeat-sprint training is a valuable HIIT weapon for combat sports and primarily used to hit target type 4. It is widely accepted that repeat sprints promote the ability to tolerate high levels of metabolic acidosis (i.e., H^+). A variety of methods from sport-specific bag or pad work, wrestling takedowns, or grappling transitions can be used, to more generic sprint intervals on a track or nonmotorized treadmill (e.g., $12 \times \geq 30$ sec-on/30 sec-off). It is important to note that primary adaptations occur in the muscles that actually perform the work, so when planning the use of repeat sprints exercise selection should be taken into consideration.

Anaerobic lactic capacity is essential for fighting styles that require repetitive high-power output for short to moderate durations (e.g., boxing, wrestling). As such, repeat-sprint HIIT is an important strategy for combat sports with repeated explosive epochs of activity or a high effort:pause ratio. Common work intervals include 15 to 40 s of maximal intensity exercise, with no more than 10 to 30 s of rest. Negative work:rest ratios of 20 sec-on/10 sec-off, as popularized by Tabata et al. (50), are also popular among combat athletes. The ability to maintain repetitive explosive efforts under fatigue is a key factor in the later rounds of combat bouts, along with recoverability between rounds. For these reasons repeat-sprint HIIT is used to increase lactic ATP production (i.e., lactate power) and promote better buffering mechanisms to improve *lactic tolerance*.

Sprint Interval Training

Sprint intervals are associated with energy contributions via alactic metabolism (target 5) with large amounts of neuromuscular fatigue (e.g., maximal intensity striking combinations or bike sprints). All combat disciplines require explosive anaerobic power to be expressed at critical points during a fight, be it to throw punches and kicks, go for a takedown, improve a ground position, or finish a fight with a submission. Alactic power can improve up to 30% by increasing the enzymes of alactic energy production, elevating phosphocreatine storage capacity, and increasing the strength of fast-twitch muscle fibers (33, 38). Surprisingly, using sprint intervals to train the alactic system to produce ATP at

extremely fast rates is also closely associated with oxidative metabolism as aerobic pathways support the replenishment of phosphocreatine reserves (e.g., $4 \times 3 \times \leq 30$ sec-on with 60 to 90 s rest between repetitions and ~180 s rest between sets). Sprint intervals elevate the magnitude of anaerobic power and are therefore commonly performed out of competition when athletes can be exposed to high levels of neuromuscular fatigue. This modality of HIIT can also be used sparingly in the later stages of fight camps as a mechanism to stimulate higher-order motor units, but it is important that volume load is reduced significantly.

Fight-Based Training

Fight-based HIIT drills, combat sport's version of GBHIIT, are a well-established approach to metabolic conditioning in all combat sports, with simulated rounds, technique drills, or situational fighting used to develop both aerobic and anaerobic metabolism (see table 11.3). Fight-based training adds an additional layer of complexity to the physical demands of long-interval HIIT, in that situational and scenario-based fighting drills concurrently challenge decision-making, the need to reactively execute a variety of offensive and/or defensive techniques, as well as the potential to change intensity level during any given repetition (i.e., moving from controlled clinch work to rapid flurries of striking). Fight-based training is largely specific to each respective combat discipline, as it often involves the technical skills discrete to that sport. For example, MMA fighters often train against multiple sparing partners and rotate 1 min epochs of takedown defense, grappling techniques, and striking for 5 min rounds (i.e., shark tank drill). Fight-based training strategies can be used to isolate specific techniques and see athletes placed under duress for the duration of an interval. A good example of this can be found in judo and wrestling, where specific throwing techniques (e.g., leg sweeps, hip throws) are executed repeatedly at high speed, with athletes looking to perform the maximal number of throws possible in a given time period. An important consideration when adopting fight-specific HIIT is that mechanical and metabolic fatigue impacts the quality of fighting techniques, and executing technical skills while fatigued can have detrimental effects on skill development. Consequently, when considering fight-based training methods, athletes should consider the effect that HIIT will have on technical as well as physiological responses.

Table 11.3 Fight-Based High-Intensity Interval Training (HIIT) Workouts Using Fight-Specific Training Methods

These short-interval HIIT workouts focus on the development of physiological attributes associated with target types 1, 2, 3, and 4.

	Striking (Muay-Thai)	Wrestling (wrestling)	Grappling (BJJ)	Hybrid (MMA)
Target	↑ alactic energy production and power output of FT fibers ↑ peak blood lactate accumulation ↑ oxidative process of recovery	↑ oxidative capabilities of ST and FT fibers ↑ cardiac output, ↑ mitochondrial proliferation, and ↑ oxygen utilization	↑ oxidative capabilities of ST and FT fibers ↑ alactic power and power output of FT and ST fibers ↑ lactic ATP energy production	↑ peak blood lactate accumulation ↑ lactic power and capacity, and ↑ maximal rate of ATP production ↑ oxidative process of recovery
HIIT type	4	1	2	3
Modality	Counter sparring drill	Endurance work	Fight gone bad drill	Shark tank conditioning drill
Format	Fighter is positioned on the ropes or in a corner. Working with a partner, the fighter must throw counter combinations against pressure.	Working with a partner, fighter drills technical work in a continuous semisparring fashion for extended work periods.	Fighter deliberately starts the drill in a compromised tactical position, then works to escape from poor defensive positions. Reset start position following each escape	One fighter competes against five other fighters, rotating between 1 min of boxing, bottom-position BJJ, Muay-Thai, take-down defense, and clinch work for the period of a 5 min interval
Intensity	High (80% to 90% HR _{max})	Moderately high (80% to 95% HR _{max})	Moderately high (80% to 95% HR _{max})	Near maximal (90% to 100% HR _{max})
Volume	4 to 8 × (3 × 60 s), 30 s recovery between sets	15 to 30 min	5 to 10 min	3 to 5 × 5 min (1 min epochs of different MMA techniques)
Work:Rest	3:1 to 1.5	Continuous work	Continuous work	5:1

Metabolic Responses to Changes in Ring Dimensions

Competition areas in combat sports vary, and can range between 5.5 m (18 ft) boxing rings to 9.1 m (30 ft) octagons. Whereas the shape and dimension of the competitive areas differ, there is little evidence to suggest metabolic demands are impacted significantly based on the size of the competition area. Indeed, it is not uncommon for boxing rings, wrestling mats, and MMA octagons to be found in a host of different regulation dimensions for competition. However, as combat sports are largely contested between two athletes in close proximity, there is little requirement to move over significantly large distances. Consequently, unlike expansive court- and field-based sports such as handball and soccer, where changing the pitch dimensions can drastically affect the physiological demand, such responses are equivocal in combat sports. Interestingly, finish-

ing rates are found to differ in MMA based on octagon dimensions, with a 48% finishing rate in 9.1 m (30 ft) octagons and 60% in 7.6 m (25 ft) octagons. However, there is little supporting evidence to indicate that these differences are consequent to altered metabolic demand.

Locomotor Responses to Changes in Ring Dimensions

Locomotor responses to changes in ring dimension are observed. In particular, the shape of the competition area, be it a circle, square, or octagon, can ultimately impact fight tactics, which in turn influences locomotion. For example, octagons and circles don't have corners, whereas boxing rings do. From a tactical perspective, boxers and Muay Thai fighters can use specific movement patterns to trap an opponent in a corner in order to gain a more dominant attacking position. In an octa-

gon there is greater opportunity to circle and move around the competition area without becoming trapped. Changes in movement patterns then support these tactical considerations be it more sideways movement or aggressive forward pressure, etc. Furthermore, an important consideration in most hybrid sports is that locomotor patterns are constantly changing as athletes transition between different fighting styles. The energetic requirements of boxing and striking movements, clinch work in wrestling, or the various postures achieved during ground fighting all present different energetic demands. Boxing and striking largely hit type 2 targets; wrestling challenges targets types 3, 4, and 5; while grappling targets HIIT types 1 and 3.

Longitudinal Training Effects of HIIT

While cross-sectional data detailing the performance standards of various combat sports, includ-

ing karate, taekwondo, boxing, and MMA, has been presented (7, 12-14, 31), understanding of the longitudinal effects of HIIT in combat disciplines is very limited. Indeed, there is no published data detailing the influence of specific HIIT interventions in combat sports. This is largely due to the fact that HIIT is rarely employed in isolation in combat sports but instead is combined with traditional fight-training modalities (e.g., sparring). Furthermore, combat sports differ greatly from field and invasion games, where HIIT modalities have traditionally been utilized. This makes it hard to fully determine the impact of HIIT on overall fighting performance, as it can be unclear as to which stimuli is causing any training adaptations (i.e., HIIT or fight training). Figure 11.6 indicates longitudinal changes in metabolic performance of world-level Olympic taekwondo fighters.

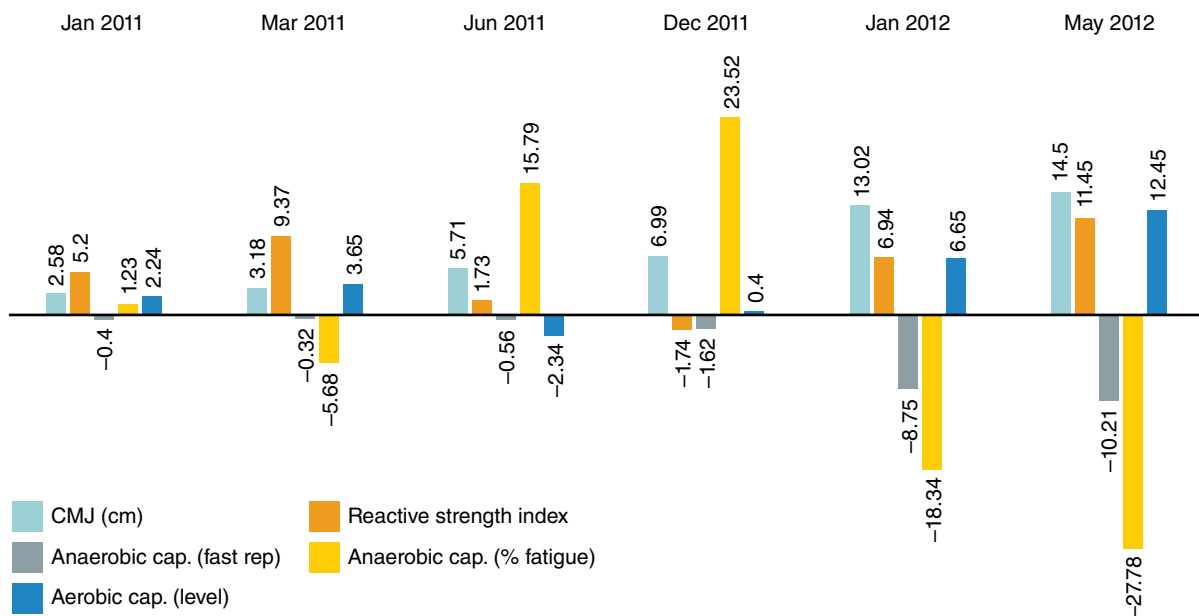


Figure 11.6 Percent change in performance standards for selected physiological measures in elite international taekwondo players preparing for the 2012 Olympic Games. These data are consequent to the distribution of HIIT represented in figure 11.3. Note, the London Olympic Games were in August 2012. Countermovement jump (CMJ) height (cm) was used as an assessment of lower-body power, while reactive strength index was measured as a proxy of explosive strength (calculated as flight time [ms]/contact time [ms] during a 40 cm drop jump). A 300 m shuttle test (12×25 m) was performed twice with a 3 min rest between efforts and used to estimate anaerobic capacity. During the 300 m shuttle, the fastest repetition was measured and the percent of fatigue between the fastest and slowest repetitions was determined. Aerobic capacity was monitored using the YoYo Endurance test (level 2).

Load Surveillance and Monitoring Tools

The relationship between *intrinsic* physiological stress and the *extrinsic* time-motion characteristics (see chapter 8) is a representation of *training load*. Presently there is a dearth of information reporting chronic training load characteristics in combat sports, and load surveillance is a relatively new concept to the fight community. As described in chapter 8, calculating session-RPE (rating of perceived exertion [RPE] $\times$ session duration) is a cost-effective way to monitor intrinsic training load (15). Importantly, session-RPE can be aggregated from multiple training modalities, be they combat-specific drills or strength and conditioning activities, to give a holistic overview of total load (see figure 11.7). This can be particularly valuable when monitoring the collective contributions of training techniques in hybrid sports. Elsewhere, heart rate monitoring represents a common method to quantify internal load. By reporting heart rate variables (e.g., max or mean beats/min) against training duration, it is possible

to establish a training impulse (TRIMP) as a proxy of training load (24).

Current advances in technology (e.g., GPS) have revolutionized extrinsic load monitoring in sports such as soccer, rugby, and Australian rules football. Interestingly, the use of wearables, inertial sensors, and tracking technologies remains in its infancy in combat sport. Stand-up fight disciplines such as boxing, karate, and taekwondo lend themselves to potentially using wearable technologies to provide crude biomechanical assessment of strike count, velocities, and power. Indeed, monitoring like this can give awareness of how training stimuli impact performance standards. However, for ground-fighting disciplines (e.g., judo, wrestling, jiu-jitsu, MMA) the monitoring of extrinsic load using inertial sensors placed on the body remains a challenge. This is largely due to the fact that reliability can be affected by opponents pulling on monitors, the sensors becoming dislodged during grappling and wrestling, or the potential for injury if athletes land on a sensor during a throw or takedown.

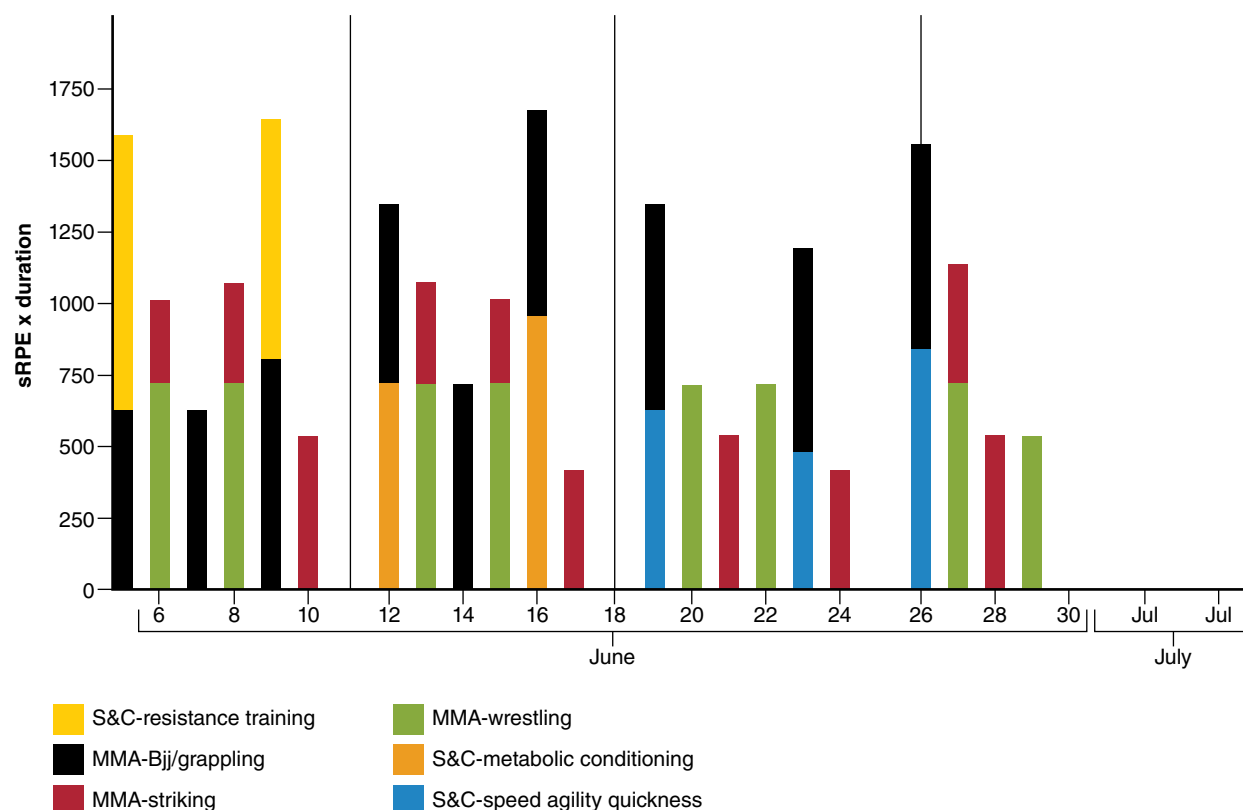


Figure 11.7 One-month training load surveillance in a professional MMA athlete using session-RPE (sRPE). Note, sRPE can be used to aggregate different types of training sessions on a single day, in order to give a holistic overview of total training load exposure.

Training Status Surveillance and Monitoring Tools

Tools to monitor training status in combat sports are often limited to noninvasive cost-effective solutions. For example, wellness monitoring using simple questionnaires, such as those described in chapter 9, gives valuable insight into an athlete's perception of health, well-being, and his ability to tolerate the training load (46). Wherever possible, wellness monitoring should be associated with an objective assessment of readiness to train (RtT). The concept of RtT is adopted by performance specialists as a reflection of intrinsic physiological status. For example, monitoring heart rate variability (HRV; see chapter 9) using technologies such as Omegawave and Morpheus provides insight into sympathetic and parasympathetic activity, with the understanding that training overload can suppress aspects of neurological activity (chapter 7). HRV has been shown to be an effective tool to detect physiological sensitivity to training stimuli and to understand if athletes are able to accommodate a specific training load or if there is suppression of physiological systems (8, 37). Other methods used as a proxy of RtT can include simple jump testing diagnostics (e.g., countermovement and/or drop jumping). High-impulse activities like jumping require rapid neuromuscular drive, and changes in peak jump height or a reactive strength index (RSI; [flight time/contact time] during a drop jump) can be indicative of suppressed neuromuscular capabilities consequent to heavy training (16, 22). Some combat sports also have punch meters that can be used as a crude assessment of training status. Monitoring sport-specific parameters such as striking force, power, and/or velocity can give valuable insights as to how performance standards are impacted by longitudinal training strategies.

Efforts to undertake longitudinal surveillance of training load are important, as understanding acute:chronic workload variation over time can be useful (28). Acute workload is defined as the absolute workload performed in 1 wk, whereas chronic workload is the 4 wk average or exponentially weighted moving average. Murray et al. (42) have indicated that accentuated spikes in acute:chronic load should be avoided, as they are associated with heightened injury incidence. Longitudinal monitoring of the acute:chronic ratio, in association with session-RPE and other intrinsic and extrinsic monitoring tools, provides invaluable surveillance insights that can be used to regulate HIIT prescription.

Strategies for Structuring the Training Program Using HIIT

Optimal physiological adaptations do not just come from the use of effective training methods but more importantly the application of the right methods at the right time and in the right way (33). Organizing the distribution of HIIT effectively between out-of-competition and fight camp training is important. Indeed, the correct sequencing of HIIT stimuli throughout the competition year is critical in order to ensure optimal development of both metabolic and neuromuscular systems. Furthermore, consideration must be given to the residual effects of training, be they acute, accumulated, or delayed (33), as poorly managed fatigue and excessive overload can present diminishing returns in response to poorly managed training.

Controllable and Uncontrollable Factors

The greatest challenge to the use of HIIT in combat sports is often the need to navigate around injuries and biomechanical accommodations (e.g., limited shoulder range of motion) that impact an athlete's capacity to execute drills in a desired fashion. Owing to the daily rigors of fight preparation, combat athletes are susceptible to injuries that require either minor adjustments in exercise selection and duration or whole-scale changes due to more severe contraindications (e.g., concussion). In addition to unforeseen injury, without a clearly defined competition structure, as found in some combat sports, it can be challenging to gain awareness for longitudinal periodization of training. Indeed, for many professional combat athletes who rarely have set annual structures but negotiate their addition to fight cards in a random fashion, it can be challenging to consistently plan the distribution of HIIT training blocks throughout the year.

Program Structure and Progression

For periodization purposes, combat athletes can be considered either out of competition or in fight camp. In those sports with set competition schedules, out-of-competition training can potentially be further periodized into a *general preparation* or *accumulation* phase, during which general fitness and strength are built, and a *specific preparation* or *transmutation* phase, in which general abilities are transferred to

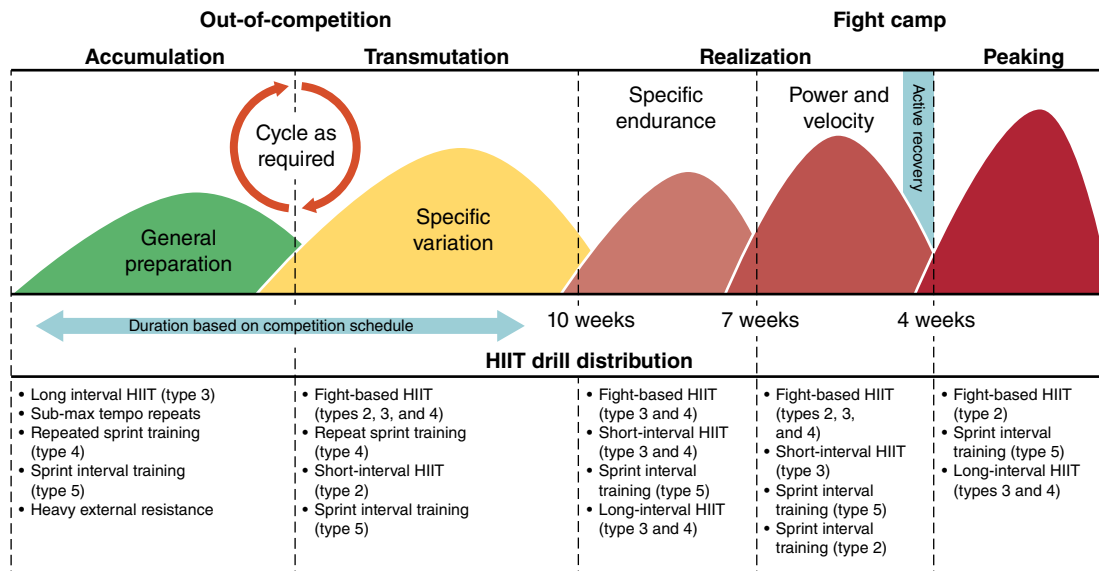


Figure 11.8 An example of the correct sequencing of training blocks and associated distribution of high-intensity training stimuli during out-of-competition and fight camp training phases for combat sport athletes.

special abilities (29). Fight camps represent the *peak-ing* or *realization* phase when athletes become acutely prepared for competition (see figure 11.8). Sadly, for the majority of combat athletes, the fight camp mesocycle is traditionally the only phase given specific attention regarding clear structure and progression, as athletes focus on an 8 to 12 wk process of reducing body weight for the official weigh-in and optimizing performance for the fight or tournament. However, performance standards within a fight are limited by the training athletes complete between fights, not just the training done in the weeks leading up to it. Working around the limitations of knowing a competition structure or not, a comprehensive training plan for combat sports needs to include out-of-competition phases that systematically improve the level of physical preparedness throughout the year. Distribution of training stimuli should be managed annually rather than just within an 8 to 12 wk fight camp, thereby giving adequate time for specific systemic adaptations to occur. This reduces the desire to use fight camps as the *only* phase to address all aspects of conditioning. Indeed, by utilizing longitudinal periodization and the concept of *phase potentiation* (23), combat athletes can elevate their performance standards beyond those achieved during 8 to 12 wk fight camps alone.

Out-of-competition phases (off-camp) should cycle between *accumulation* and *transmutation*, during

which fighters need to prioritize the progression of structural and functional adaptations that enhance aerobic, lactic, and alactic metabolic power. These complex physiological adaptations (e.g., mitochondrial biogenesis, H^+ buffering capacity, oxidative enzyme kinetics, myocardial enlargement) are the foundations on which fight-specific fitness is optimized, and in many cases the realization of these adaptations requires months to become apparent. Philosophically, out-of-competition training is about “putting money in the physiological bank” so that the value of these savings can be returned during fight camp.

When an athlete finally enters a fight camp phase, the focus is *realization* of metabolic efficiency specific to the characteristics of the fight (figure 11.8). This differs relative to each combat discipline (i.e., effort:pause ratio), and for hybrid sports such as MMA it can also be affected by the style of opponent (e.g., “predominantly a striker,” “strong ground game”) or the way an athlete feels he can dominate a fight. Fight camps themselves should cycle training blocks that emphasize the progression of physiological characteristics such as local neuromuscular endurance, cardiovascular capacity, and high-intensity speed (33), so that at the end of the 8 to 12 wk camp athletes can express very high-intensity epochs of neuromuscular power with the metabolic capacity to replicate these efforts continuously over time.

Incorporation of HIIT

In a combat athlete's out-of-competition training, accumulation and transmutation phases should be cycled for 3 to 8 wk. A week of de-loading would preferably separate the phases, and as soon as clarity around a fight schedule becomes apparent, appropriate planning to cease out-of-competition cycles should be planned for 8 to 12 wk prior to a fight. The priority of an accumulation phase is to develop maximal power capabilities and a general level of conditioning in all neuromuscular and metabolic systems (figure 11.8). This includes maximizing the contractile properties of fast- and slow-twitch fibers, developing glycolytic capacity and H^+ tolerance, and elevating the rate at which the body can process energy via oxidative metabolism, i.e., $\dot{V}O_{2peak}$ (29, 33, 38). The most appropriate HIIT modalities for the accumulation block include:

- High-intensity long intervals (cardiac output)—types 3 and 4
- Submaximal tempo repeats (oxidative capacity)—types 1 and 3
- Repeat-sprint intervals (lactate tolerance)—types 4 and 5
- Sprint intervals (neuromuscular power)—type 5
- And, outside of HIIT, heavy resistance work (developing contractile properties; type 6)

A transmutation or transition phase includes more sport-specific training factors and a reduction in general preparatory activities. Essentially, the general physical abilities developed in the accumulation phase are *transmuted*, or transitioned, into event-specific readiness. In some nomenclature the transmutation phase is also referred to as a *special preparation phase*. The focus of the transmutation phase is to emphasize the neuromuscular capabilities of specific muscle groups, challenge the metabolic demands of competition, and promote recoverability through enhanced regenerative qualities of the oxidative system. The selection of appropriate HIIT modalities is critical in a transmutation phase, so as to realize the potential of the general

training that has been completed before, but not to overload specific attributes that could potentially lead to detrimental fatigue (see figure 11.8). HIIT strategies would include:

- Fight-based training (cardiac output)—types 1, 2, 3, and 4
- Repeat-sprint intervals (lactate capacity)—types 4 and 5
- Short intervals (alactic power)—type 2
- Sprint intervals (neuromuscular power)—type 5

Finally, when an athlete moves from out-of-competition conditioning to fight camp, the focus shifts to specific conditioning that allows an athlete to make an 8 to 12 wk progression to competition standards. Much of the training done during this block is focused on fight-based HIIT, including sparring, simulated rounds, and high-intensity technique work. Within fight camp, athletes should cycle through shorter training blocks of 2 to 3 wk, moving progressively from high-volume capacity work to high-intensity explosive efforts and active recovery. The fight camp (e.g., effort:pause ratio) should build on out-of-competition adaptations and maximize their influence on fight-specific skills. HIIT modalities should include:

- Fight-based training (lactic capacity)—types 3, 4, and 5
- Short intervals (lactic power)—types 3 and 4
- Short intervals (alactic power)—type 2
- And, outside of HIIT, speed-strength resistance work (developing neural activation)
- Active recovery

Sample Training Programs

Tables 11.4 and 11.5 provide sample HIIT programs used in various combat sports during different phases of training, including both out of competition and fight camp. Each of these programs has very specific physiological objectives, and they demonstrate how HIIT is a central aspect of the programming strategy in combat sports.

Table 11.4 Weekly Plan for an Out-of-Competition Accumulation Phase in Elite Olympic Taekwondo Players

	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
0700							OFF
0800		0800-0930 S&C (Type 3) Long-interval HIIT	0800-0930 TKD (Type 4) High volume striking			0800-0930 S&C (Type 5) Repeat sprint intervals and lactate tolerance work	
0900	0900-1030 TKD (Type 1) Technical drilling and technique volume				0930-1100 TKD (Type 1) Technical drilling and technique volume		
1000							
1100		1130-1330 TKD Technique work in the clinch	1130-1300 S&C Heavy resistance training (Type 6)	1130-1330 TKD (Type 3) Movement and agility to scoring chances			
1200							
1300							
1400	1400-1530				1400-1530		
1500	TKD (Type 2) Speed/high tempo striking				TKD (Type 2) Speed/high tempo striking		
1600				1630-1800			
1700		1730-1830 TKD Individual technique work		TKD (Type 4) Sparring			
1800	1800-1900 S&C Power emphasis resistance training (Type 6)				1800-1900 S&C Heavy resistance training (Type 6)		
1900							

TKD=taekwondo-specific workout; S&C=supplementary strength and conditioning workout

Table 11.5 Weekly Plan for the Fight Camp Phase of a Professional MMA Fighter

	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
0700							OFF
0800							
0900							
1000							
1100	1100-1300 Grappling/BJJ (Type 3) Long-interval HIIT			1100-1300 Muay-Thai/Striking (Type 2) Speed/high-tempo striking and short-interval HIIT	1100-1300 Grappling/BJJ (Type 1) Technical drilling and long-interval HIIT	1600-1800 MMA Sparring (Type 5) Repeat intervals and lactate tolerance work	
1200		1100-1300 Wrestling (Type 3) Technical drilling and long-interval HIIT					
1300							
1400							
1500							
1600	1600-1800 S&C (Type 6) Heavy resistance training	1600-1800 Muay-Thai/Striking (Type 2) Explosive speed and max power short interval HIIT	1600-1800 MMA Sparring (Type 4) Explosive speed and max power short-interval HIIT	1600-1800 Wrestling (Type 3) Anaerobic capacity and lactate tolerance work	1600-1800 S&C (Type 6) Power-emphasis resistance training		
1700							
1800							
1900							

BJJ = Brazilian jiu-jitsu; MMA = mixed martial arts; S&C = supplementary strength and conditioning workout

CASE STUDY

Shortly following the 2012 London Olympics, I became the head strength and conditioning coach for the Great Britain Olympic Taekwondo team, with the remit of directing the team's physical preparations for the Rio Olympics in 2016. Expectations were high with the team having achieved its most successful Olympic Games ever in London, winning gold and bronze medals. Sadly, however, London 2012 was a frustrating experience for Bianca Walkden, who was the team's up-and-coming +67 kg heavyweight. Two years prior to the London Games, Bianca agonizingly ruptured the ACL in her left knee, an injury that effectively ended her chances of securing a place to compete at London 2012.

Fast-forward 2 yr, and fueled by her desire not to miss out on another Olympic Games, Bianca was committing all her energies to becoming the best fighter she could be. An absolute coach's dream, Bianca gave everything to her physical preparation. Things were coming together, her physical standards were increasing exponentially, and in 2014 Bianca secured her maiden European crown at the European Championships in Baku, Azerbaijan. Two months later, tragedy struck again! At the world Grand Prix in Turkey, Bianca was dominating her opponent with powerful leg kicks, but after landing awkwardly she fell to the ground clutching her left knee. Heartbroken, she knew straightaway what it was, and she couldn't believe it was happening with the next Olympics less than 2 yr away. The ACL in her left knee had ruptured again.

After surgery Bianca regrouped and targeted the 2015 World Championships in Chelyabinsk, Russia, as her comeback in a major championship. This was only 8 mo away! As a support team we felt this was ambitious given that knee injuries in a kicking sport like taekwondo can significantly affect performance. However, we set about her rehabilitation and I dedicated my efforts to progressively increasing her physical capabilities to compete again at the most elite level. We built out the physical development plan alongside her rehabilitation needs and the coaches' integration of taekwondo-specific drills. With 8 mo, we periodized the steady reconditioning of her bioenergetics through a variety of continuous and HIIT modalities. Progress didn't happen as planned, and we were at a loss to understand why. Local muscular endurance was improving, and metabolic conditioning was getting better, but her repaired knee was still an issue and things weren't right. After much consultation, the decision was made by the medical team to go back into her knee 2 mo after the initial ACL surgery, make minor surgical repairs, and hope that healing would be accelerated as a consequence. Eight months of preparation time for the Worlds crazily became 6 mo, the first two of which she would be very limited in any sport-specific capabilities. We all felt like this was an impossible task to get her to the Worlds, but with crucial Olympic ranking points for Rio on the line it was almost nonnegotiable that she had to fight, otherwise her dreams of competing at the Olympics would once again be ruined.

Immediately the periodization of metabolic conditioning changed significantly. We moved to four HIIT sessions a week, two of which were focused on neuromuscular and metabolic power, and two that truly challenged her glycolytic system and the ability to tolerate high levels of metabolic acidosis. Many of the early sessions had to be performed in nontraditional fashions as we attempted to accelerate the metabolic conditioning, all the time working around the injured knee. Adopting a reverse-periodization strategy, the overarching focus became the development of the highest levels of anaerobic and aerobic power possible, and then use of HIIT strategies to build capacity/endurance against newly found levels of performance.

Only 6 mo following the second surgery on her torn ACL, Bianca was walking out to compete at the World Championships in Russia. It was truly remarkable, and knowing the dedication she had given to the rigorous HIIT schedule we prescribed, it was truly one of the most rewarding moments of my career. I never thought we would get her there, but with a clearly defined HIIT plan and unwavering commitment, Bianca made it to the Worlds! She

(continued)

CASE STUDY (CONTINUED)

was operating with about 80% neuromuscular power in her left leg, but her glycolytic and oxidative conditioning levels were the highest they had ever been. It was a punt, but she was willing to take the chance against the world's best.

At great surprise to everyone who knew what Bianca had been through, she began to progress through the tournament, winning through the early rounds and ultimately making it to the final round. In the final, she was going up against a French athlete who was recognized as one of the most powerful and dominant female fighters of recent times. Following a crazy fight, much to everyone's surprise, Bianca won!! Eight months after an ACL rupture, 6 mo after a secondary knee surgery, and essentially following 4 months of high-intensity training, Bianca had become world champion!! It was a remarkable turnaround, and for me it reiterated how valuable it is to have a clearly defined training plan that is aligned to the specific needs and demands of competition.

The story didn't end there! Bianca immediately got back to work, continuing to work on her physical conditioning. Amazingly, 12 mo after becoming world champion she retained her European title and was the number one ranked fighter in the world. She had amassed enough ranking points in half the amount of time to other competitors to confirm her place at the Rio Olympic Games. She took home a Bronze medal at the Olympic Games and was somewhat disappointed. But this was an amazing achievement for an athlete who had won two European championships, a world championship, and a Bronze medal at the Olympic Games just over 2 yr after a potential career-ending knee surgery.